QUANTITATIVE METHODS

MODULE 1: RATES AND RETURNS

I. SUMMARY

Unit 1: Explain interest rate as the sum of a real risk-free rate and premiums

Unit 2: Measurements for wide range of assets and time periods

Unit 3: Measurements for special assets

II. QUESTIONS

Unit 1: Explain interest rate as the sum of a real risk-free rate and premiums

Question 1:

Which of the following risk premiums is most relevant in explaining the difference in yields between 30-year bonds issued by the US Treasury and 30-year bonds issued by a small private issuer?

A. Inflation

B. Maturity

C. Liquidity

For Activity 2.1: Self-practice	**For Activity 2.2: Self-review**
Note down short explanations for your answer choices.	*Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.*

Please note down here

Unit 2: Measurements for wide range of assets and time periods

Question 1:

An asset manager's portfolio had the following annual rates of return:

Year	Return
20X7	+6%
20X8	–37%
20X9	+27%

The manager states that the return for the period is –5.34%. The manager has reported the:

A. arithmetic mean return.

B. geometric mean return.

C. holding period return.

For Activity 2.1: Self-practice
Note down short explanations for your answer choices.

For Activity 2.2: Self-review
Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.

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Question 2:

A bond was purchased exactly one year ago for $910 and was sold today for $1,020. During the year, the bond made two semi-annual coupon payments of $30. What is the holding period return?

A. 12.1%.

B. 18.7%.

C. 6.0%.

For Activity 2.1: Self-practice	**For Activity 2.2: Self-review**
Note down short explanations for your answer choices.	*Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.*

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Question 3:

Time-weighted returns are used by the investment management industry because they:

A. take all cash inflows and outflows into account using the internal rate of return.

B. result in higher returns versus the money-weighted return calculation.

C. are not affected by the timing of cash flows.

For Activity 2.1: Self-practice	**For Activity 2.2: Self-review**
Note down short explanations for your answer choices.	*Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.*

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Question 4:

Which of the following statements is most accurate? The money-weighted return:

A. ignores cash withdrawals and additional cash investments.

B. measures what the investor actually earned on the funds invested.

C. should be used to compare the performance of different investment managers.

For Activity 2.1: Self-practice	**For Activity 2.2: Self-review**
Note down short explanations for your answer choices.	*Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.*

Please note down here

Question 5:

An investor buys a share of stock for $40 at time t = 0, buys another share of the same stock for $50 at t = 1, and sells both shares for $60 each at t = 2. The stock paid a dividend of $1 per share at t = 1 and at t = 2. The periodic money-weighted rate of return on the investment is closest to:

A. 22.2%.

B. 23.0%.

C. 23.8%.

For Activity 2.1: Self-practice	**For Activity 2.2: Self-review**
Note down short explanations for your answer choices.	*Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.*

Please note down here

Unit 3: Measurements for special assets

Question 1:

A security portfolio earns a gross return of 7.0% and a net return of 6.5%. The difference of 0.5% most likely results from:

A. inflation.

B. fees.

C. taxes.

For Activity 2.1: Self-practice	**For Activity 2.2: Self-review**
Note down short explanations for your answer choices.	*Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.*

Please note down here

Question 2:

The value of an investment increases 5% before commissions and fees. This 5% increase represents:

A. the investment's net return.

B. the investment's gross return.

C. neither the investment's gross return nor its net return

For Activity 2.1: Self-practice	**For Activity 2.2: Self-review**
Note down short explanations for your answer choices.	*Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.*

Please note down here

Question 3:

The strategy of using leverage to enhance investment returns:

A. amplifies gains but not losses.

B. doubles the net return if half of the invested capital is borrowed.

C. increases total investment return only if the return earned exceeds the borrowing cost

For Activity 2.1: Self-practice	**For Activity 2.2: Self-review**
Note down short explanations for your answer choices.	*Compare your choices with the final answer and correct (if needed). Note down your own key takeaways/lessons learned.*

Please note down here